

Mathematics of Vacuum Tape Simulation

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1 Introduction

This document formalizes the digital signal processing models to be implemented in the **Vacuum Tape Sim** plugin. These models define the non-linear transfer functions of vintage vacuum tube power supplies and magnetic tape systems.

2 Vacuum Tube “Voltage Sag” Compression

The virtual tube models a power supply that “struggles” under load. Heavy transients cause the available supply voltage to drop, creating program-dependent compression.

2.1 Current Drain and Recovery (Voltage State)

To prevent the voltage from rippling at the audio rate, the absolute input $|x[n]|$ is first smoothed by a one-pole envelope follower to estimate the energy load $e[n]$:

$$e[n] = e[n-1] + \alpha_{env} \cdot (|x[n]| - e[n-1]) \quad (1)$$

The system maintains an internal state variable $V[n] \in [0.0, 1.0]$, representing the available voltage. To ensure smooth, mastering-grade compression without low-frequency ripple, the system uses a **True RMS Detector** with a **Dual-Stage Recovery** and a **Soft-Knee** threshold.

1. True RMS Detection (Ripple Rejection): Instead of tracking peak absolute values, the energy is squared and passed through a 2nd-order (two-pole) low-pass filter to aggressively reject low-frequency ripple while preserving fast transient reaction:

$$x_{sq}[n] = x[n]^2 \quad (2)$$

$$e_1[n] = e_1[n-1] + \alpha_{env} \cdot (x_{sq}[n] - e_1[n-1]) \quad (3)$$

$$e_{rms}[n] = e_{rms}[n-1] + \alpha_{env} \cdot (e_1[n] - e_{rms}[n-1]) \quad (4)$$

(Using $\tau_{env} \approx 5ms$).

2. Soft-Knee Drain: To prevent sharp discontinuities when crossing the threshold T , the drain rate uses a squared soft-knee polynomial:

$$Drain[n] = \alpha_{drain} \cdot \max\left(0, \sqrt{e_{rms}[n]} - T\right)^2 \quad (5)$$

3. Dual-Stage Program-Dependent Recovery: A secondary, slow envelope $e_{slow}[n]$ tracks sustained energy to dynamically adjust the recovery time. If the audio is sustained (like a bassline), recovery is slow ($\tau_{slow} \approx 500ms$). If it is a fast transient (like a snare), recovery is fast ($\tau_{fast} \approx 50ms$):

$$e_{slow}[n] = e_{slow}[n-1] + \alpha_{slow_env} \cdot (e_{rms}[n] - e_{slow}[n-1]) \quad (6)$$

$$\beta_{dynamic}[n] = \beta_{fast} \cdot (1 - e_{slow}[n]) + \beta_{slow} \cdot e_{slow}[n] \quad (7)$$

4. Voltage Update:

$$V[n] = \max(0.1, V[n-1] - \text{Drain}[n] + \beta_{\text{dynamic}}[n] \cdot (1 - V[n-1])) \quad (8)$$

All coefficients (α, β) are calculated using the standard time-constant formula $\alpha = 1 - \exp(-1/(\tau \cdot F_s))$.

2.2 Dynamic Gain

The output signal $y[n]$ is the input scaled by the available voltage:

$$y[n] = x[n] \cdot V[n] \quad (9)$$

3 Magnetic Hysteresis (Langevin Model)

To physically model magnetic tape particle saturation, we replace the standard tanh waveshaper with the **Langevin Function**, which describes anhysteretic magnetization.

3.1 The Langevin Function

The Langevin function $L(x)$ is defined as:

$$L(x) = \coth(x) - \frac{1}{x} \quad (10)$$

For numerical stability near zero, the Taylor series approximation $L(x) \approx x/3$ is used when $|x| < 10^{-4}$.

3.2 Stateful Hysteresis and Asymmetry

To capture both magnetic lag and even-harmonic generation (asymmetry), the magnetization $y[n]$ tracks the continuous rate of change of the input. We normalize the discrete derivative $\Delta x[n] = (x[n] - x[n-1]) \cdot F_s$ to ensure sample-rate independence:

$$y[n] = L(G \cdot (x[n] + x_{\text{bias}}) - h \cdot L(\beta \cdot \Delta x[n])) \quad (11)$$

Where:

- G : Drive (Scale factor for input gain).
- x_{bias} : A subtle DC offset to introduce asymmetry (even harmonics).
- h : Coercivity (Amount of magnetic lag/memory).
- β : Smoothing factor.

3.3 Virtual Bias (Macroscopic Approximation)

Instead of processing an ultrasonic sine wave (which requires 16x oversampling), the macroscopic effects of the tape bias calibration are modeled via a user parameter $B \in [-1.0, 1.0]$, where 0.0 is perfect calibration.

The Bias parameter dynamically modulates the Hysteresis smoothing (β) and the High-Frequency Gap Loss cutoff (f_c):

- **Smoothing (β):** $\beta(B) = \beta_{\text{base}} \cdot 2^B$. Over-biasing ($B > 0$) softens the transients, while under-biasing ($B < 0$) causes faster, grittier saturation.
- **High-Frequency Shift:** The gap loss cutoff frequency shifts inversely: $f_c(v, B) = f_c(v) \cdot 2^{-B/2}$. Over-biasing darkens the tone, while under-biasing brightens it.

4 Pre/Post-Emphasis EQ (NAB vs. CCIR)

To combat noise and limitations of magnetic particles, real tape machines apply a pre-emphasis EQ before hitting the tape (the non-linear Langevin curve), and an exact inverse de-emphasis EQ upon playback. Because the signal is saturated *between* these EQs, frequencies boosted by the pre-emphasis will distort earlier.

This is implemented by wrapping the Hysteresis process in Shelf Filters. A time constant τ maps to a transition frequency $f = \frac{1}{2\pi\tau}$.

4.1 NAB (American Standard)

The NAB standard at 15 IPS uses time constants of $3180\mu\text{s}$ (≈ 50 Hz) and $50\mu\text{s}$ (≈ 3183 Hz).

- **Pre-Emphasis:** Low-shelf boost at 50Hz, High-shelf boost at 3183Hz.
- **De-Emphasis:** Exact inverse Low-shelf and High-shelf cuts.

4.2 CCIR / IEC (European Standard)

The CCIR standard at 15 IPS uses only a $35\mu\text{s}$ (≈ 4547 Hz) time constant, leaving the low-end flat.

- **Pre-Emphasis:** High-shelf boost at 4547Hz.
- **De-Emphasis:** Exact inverse High-shelf cut.

5 Tape Playback Losses

Frequency-dependent losses are modeled using standard digital biquad filters implementing the Audio EQ Cookbook formulas. The generalized difference equation for a biquad is:

$$y[n] = (b_0x[n] + b_1x[n-1] + b_2x[n-2] - a_1y[n-1] - a_2y[n-2])/a_0 \quad (12)$$

5.1 High-Frequency “Gap Loss” (Tape Speed)

The cutoff frequency f_c is determined by the physical wavelength of the signal matching the playback head gap. As a result, the cutoff scales linearly with the physical tape speed parameter v (in inches per second, IPS), modified by the Bias parameter B :

$$f_c(v, B) = (v \cdot 650.0 \text{ Hz}) \cdot 2^{-B/2} \quad (13)$$

(Example: 15 IPS ≈ 9.75 kHz cutoff. 30 IPS ≈ 19.5 kHz cutoff).

5.2 Low-Frequency “Head Bump” (Contour Effect)

To model the geometric resonance of the playback head, a Peaking EQ filter is applied. Because this resonance is tied to physical wavelength, the center frequency shifts dynamically with tape speed v :

$$F_{bump}(v) = v \cdot 3.0 \text{ Hz} \quad (14)$$

(Example: 15 IPS ≈ 45 Hz bump. 30 IPS ≈ 90 Hz bump).

The filter uses $Q \approx 1.5$ and Gain $\approx +2.5$ dB. The coefficients map the standard amplitude variable $A = 10^{\text{gain}/40}$.

5.3 Low-Frequency Roll-Off (Faraday’s Law)

Due to Faraday’s law of induction, the output voltage of a tape playback head is proportional to the time derivative of the magnetic flux ($d\Phi/dt$). A playback head cannot induce a current from static (DC) or extremely low-frequency magnetic fields, causing a natural 6dB/octave roll-off toward 0Hz. To physically model this, a **1st-order High-Pass Filter** is applied with a cutoff frequency around 25 Hz.

6 Tape Transport (Wow & Flutter)

Mechanical instability is modeled as a modulated fractional delay line.

6.1 Mean-Reverting Wow (Ornstein-Uhlenbeck Process)

The Wow frequency follows an Ornstein-Uhlenbeck stochastic process, returning to a center value μ :

$$f_{wow}[n] = f_{wow}[n-1] + \theta(\mu - f_{wow}[n-1]) + \sigma\epsilon[n] \quad (15)$$

Where θ is the rate of mean reversion, σ is the volatility, and $\epsilon[n]$ is standard Gaussian noise.

6.2 Fractional Delay

The output is read from a circular buffer using a reading pointer $P[n]$ modulated by the LFOs:

$$P[n] = n - (D_{base} + D_{wow} \cdot A_{wow}[n] + D_{flutter} \cdot N[n]) \quad (16)$$

Sub-sample accuracy is maintained via 3rd-order Hermite interpolation.